

Climate Predictability and Variability in the Presence of Anthropogenic Climate Change: A Proposal for a Princeton Climate Prediction Center

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June 24 2000

We propose the establishment of a Center in climate prediction, involving both GFDL and Princeton University. Our ultimate goal is to develop a predictive capability for the climate system on timescales from the seasonal-to-interannual, on which the initial conditions are of paramount importance, to the multi-decadal, upon which the signal from anthropogenic climate change may begin to dominate. The activities of the Center will be designed to complement and supplement existing research at GFDL and Princeton University in such a way as to create a pre-eminent center for climate prediction, fully integrated with NOAA's emerging Climate Services.

The activities of the Center will include the continual production of ensembles of climate integrations for the next several decades, eventually incorporating the present state of the climate system as initial conditions. The integrations will be accompanied by assessments of the confidence we can place in them, and thereby assessments of the predictability of climate on interannual to multi-decadal timescales. Our predictions will be continually tested using the available observational networks, and the data and reports will continually be made available to the public and the scientific community. The accompanying scientific activities will include investigations into the mechanisms of climate variability on interannual to multi-decadal timescales, including the variability that has occurred in the past, and the interaction of natural variability with anthropogenic climate change.

1 Our Goals

The simulation and understanding of natural variability on multi-decadal scales intersects the issue of human-induced climate change in a variety of ways. It begets the question of whether predictions¹ of climate a decade or more in advance can be improved by initializing and predicting the evolution of natural variability on these time-scales. This question motivates our proposal. Our goal is the creation of a prediction/assessment capability that seamlessly connects the seasonal-to-interannual timescale to both natural and anthropogenic changes on longer time scales. Our explicit goal is to provide predictions

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¹On these long timescales, the word 'projection' is often preferred over 'prediction' because of the virtual impossibility of predicting global-scale human decisions. For simplicity we use the word prediction throughout this document, with the understanding that our predictive capability may be limited and that long-term anthropogenic climate change is not just a prediction in the usual initial-value-problem sense.

of climate and climate change over the next 50–100 years, incorporating estimates of all likely relevant future forcing terms, and initializing the state of the climate system with observed variables. The forcings will include an array of greenhouse gases, with an interactive carbon cycle, as well as statistically based estimates of the effects of future volcanic activity and alteration of the land surface, for example by deforestation. The initial states will include, in particular, the current oceanic state as observed by both *in situ* and remote instruments, including a network of floats and satellites. Our experimental predictions will ultimately be based on an ensemble of model integrations, and so will be probabilistic in nature, with a distribution of predicted states determined by the uncertainty in the initial conditions, model errors and errors in model forcing, filtered and amplified by the natural variability and unpredictability of the climate system. The quantification of climate predictability and variability will be a key element of the Center.

In summary, we seek to be the leading provider of accessible and scientifically credible climate predictions in the US, in addition to providing scientific research in support of more operational predictions elsewhere. We envision that the information we will provide will be invaluable to those performing impacts studies and those involved in policy, in addition to the climate change and IPCC communities. The Center will build on expertise at both GFDL and Princeton University. It will interact with and adjoin the existing research groups at GFDL and Princeton University, and it will interface with observational and data analysis programs elsewhere (e.g., PMEL, AOML, NODC). Overall, we intend and expect that the Center will play an integral role in NOAA's emerging Climate Services program.

2 Our Product

The primary product from such a Center will be a series of experimental climate change predictions from the seasonal-to-interannual to the multidecadal timescale, as well as research into climate predictability and development of methods for such predictions. As some of these predictions become more routine, we envision that the techniques developed will be passed on to more operational centers within NOAA. Nevertheless, for the next decade at least, predictions on the longer timescales may be performed in a research environment side-by-side with research activities into the nature of and limits to the predictability itself.

The probabilistic nature of climate change will necessitate that our predictions primarily be made in the form of changes to the probability distributions of climate. While the current generation of models is not capable of making credible predictions of climate change on the regional scale, it is expected that research conducted in this Center will facilitate the future production of such regional climate change forecasts, but the timescales on which such forecasts will be credible is not as yet determined.

Our model results, and the interpretation of these results, will be made available via the publication of periodic assessments, or reports, and periodic briefings to the interested community. The frequency of such reports will vary according to the timescale of concern, perhaps varying from an annual to a five-year period. In addition, we propose to establish a state-of-the-art web site for the dissemination of both the raw model data and the processed and interpreted data. We envision this web site as comprehensive, and of use to the general public, the private and public sectors, and the scientific community. For the general public, our goal is for the site to serve as the focal point in the U.S. for information on potential climate change. This will encompass the scientific basis of climate change, the characteristics and likelihood of potential future changes, and possible impacts. Indeed, we would like this site to be a catalyst for generating interest in the general public and the media on these issues. Similarly, we envision that both private and public sectors will benefit enormously from credible and accessible assessments of the likelihood of future changes in (for example) temperature and precipitation. Energy and water resources planning are but two obvious examples. And as our knowledge of regional climate

change improves, the benefits to these communities become correspondingly greater.

For the scientific community the web site will serve as an easy access to a wide variety of models and model-based products. A specific effort will be made to provide easy access to model output for the climate change impacts community. [Software developed at PMEL (the ‘Live Access Server’) will greatly help in this regard.] Scientific communities (such as IRI and IPCC) will also be able to access customizable subsets of model output for their own assessments of its usefulness and impact. Finally, and importantly, our scientific reports and publications in the refereed scientific literature will provide an archival and high quality foundation for all of the activities we describe.

The effort involved in all these activities and products will, of course, be substantial. However, if realized properly, it will be a valuable resource for NOAA and the nation. The following section describe the scientific problems associated with such an endeavour, and is followed by a summary of the proposed activities (see also fig. 1).

3 Climate Predictability

The predictability of any system is normally defined in terms of how well one can predict, or forecast, the future state of the system, generally compared to some prior estimate that one can obtain trivially, like persistence or climatology. The predictability of both weather and climate is limited because they are chaotic dynamical systems. Daily weather is essentially unpredictable beyond a couple of weeks, because the information in the initial conditions is largely lost and forecasts show little more skill than climatology. However, the climate system as a whole may be much more predictable than this, because many components (such as the ocean) vary much more slowly, and these provide slowly varying boundary conditions for the atmosphere. On a *daily* timescale the variability of weather is sufficiently large that a detailed knowledge of these boundary conditions provides only a very small increase in skill: essentially the signal to noise ratio is very small. However, this does not necessarily hold on longer timescales; the signal to noise ratio improves and the average weather may indeed be more predictable than a climatological prior if the state of the ocean is known. Indeed on seasonal-to-interannual timescales there is evidence that numerical forecasts can be made that improve on climatology (Anderson *et al.* 1999). Much of the source of this skill arises from predicting the conditions of the tropical ocean, and the effects this has on the global atmosphere. Of course different physical and dynamical processes dominate at the various timescales involved, and the way in which predictability is judged consequently must differ. It is useful to identify three distinct regimes:

1. The seasonal-to-interannual timescale, on which the forecast problem is largely an initial value problem. Skill resides in predicting the atmospheric response to low-latitude SST anomalies, and thus the prediction of El Niño events is particularly important. Soil-moisture and surface albedo are also important components of initial conditions that affect seasonal skill.
2. A longer, multiyear-to-interdecadal, timescale on which there is significant natural variability, likely with origins in the ocean and perhaps involving coupled ocean-atmosphere modes in the tropics and/or extra-tropics. Such long-term variability may also arise from variability of frequency and intensity of El Niño events.
3. The timescale at which the response to greenhouse gases dominates that of natural variability. This timescale is uncertain, for it depends both on the rate of emissions of greenhouse gases and aerosols, and the nature of the averaging process used to define the signal.

Effort in the community has focused largely on the first and third of these. For example, one of the goals of the Experimental Prediction Group at GFDL is to understand predictability at the seasonal-

to-interannual timescale, and the IRI at Lamont-Doherty concentrates on forecasts and their associated impacts at this timescale. The scientific problem is largely an initial value one. In contrast, in most global warming scenarios the precise initial state of the climate system has not been regarded as of first-order importance, and for centennial timescales this may in fact be appropriate. In contrast to either of these, for the decadal timescale *both* natural variability (and therefore the initial conditions of the climate system) *and* the forced response to increasing greenhouse gases can be expected to be important. It is the interaction of the initial value problem and the forced response that is central to our proposed activities. An illustration of the importance of this emerges from the simulations of Delworth and Knutson (2000). They performed an ensemble of five integrations over the 20th century, using a climate model that differed only in its initial conditions. In all five integrations there is significant natural variability, and one member of the ensemble provides a remarkably good match to the observed global temperature record (fig. 2 and fig. 3). The implication is that *both* natural variability and the forced response to increasing concentrations of greenhouse gases are important on such timescales.

3.1 Causes of unpredictability in the climate system

As in many predictability problems, lack of skill is ultimately caused by:

1. Imperfect models, including imperfect knowledge of forcing and boundary conditions.
2. Imperfect knowledge of initial conditions.

The errors from these sources are amplified by internal instabilities and feedbacks in the dynamics of the system over which we have no control. From a practical standpoint then, our overall effort will involve continually improving the numerical models we use to predict climate and initializing our models with the best possible data. More specifically, we expect that difficulties in climate prediction will arise from our lack of understanding or ability in the following areas:

1. In our ability to initialize the initial state of our climate models with a sufficiently accurate state of the ocean and sea-ice, and sufficiently accurate land-surface conditions.
2. In understanding the nature of decadal variability in the ocean–atmosphere–land–ice system. Specifically, we must better understand the nature and causes of various decadal-scale oscillations in the climate system, which in turn requires:
 - (a) Understanding and modelling the mechanisms determining the evolution of the state of the ocean (e.g., SST anomalies) and sea ice.
 - (b) Understanding and modelling the effects of such SST and sea-ice anomalies on the state of the atmosphere.

Furthermore, because we cannot realistically expect that our models and data assimilation will ever be perfect, we must quantify the nature of climate variability that has occurred in the past, from all possible causes, in order to gain an appreciation for what might occur in the future.

3. In understanding and predicting the likely future course of the radiative forcing functions of the atmosphere, including greenhouse gases, sulfur and aerosols and, conceivably, volcanic events. The variability of both the terrestrial and marine components of the carbon cycle are also major factors in this.

4. In understanding which components of the climate system are predictable, and quantifying the level of predictability we might attain. This in turn entails understanding the complex feedbacks that determine the nature of the response of the climate system to natural and anthropogenic forcing.

These issues will now each be discussed in more detail.

4 Obtaining an Initial State: Ocean Data Assimilation

On timescales beyond a month, the signal in the climate system will largely arise from the slowly varying components — primarily the ocean and sea ice, and perhaps the land surface. The dynamics of the ocean is known to contain timescales of decades (the wind-driven circulation) to centuries and even millennia (the abyssal circulation), and thus to capture any climate predictability we must initialize our climate models with the current state of the ocean. We do not know how long the influence of such initial conditions will be felt. Nor do we know well how much inertia resides in the land surface, in particular soil moisture content and albedo, but these too must ultimately become part of the initial conditions.

A number of ocean data sets exist, including the well-known Levitus data set and data emerging from the WOCE program. We also propose to take advantage of new data sets that will be emerging over the next few years, in particular the Argo float data set, and SST and sea-surface height (SSH) data from satellites. The Argo program (Argo 1998, 2000) is a follow-on to the WOCE hydrographic, float and expendable bathythermograph sampling networks, but uses an array of relatively cheap, profiling PALACE-like floats (e.g., Davis 1998). The goal of the program is to have a global array of such floats, with one float profiling to about 2000 m depth about every 10 days in every 3-degree square of the ocean, providing measurements of temperature, salinity, pressure and (from the float trajectories) a velocity field at the reference depth. All told, about 80,000 T/S profiles will be taken per year, sufficient to provide a good picture of the large-scale baroclinic structure of the upper ocean, including some information below the main thermocline. The floats do not sample the abyss, but this is both much more uniform than the upper ocean, and the typical timescales on which it varies are for the most part much longer than the decadal timescales that concern us here. Nor is the density of the floats sufficient to resolve mesoscale eddies.

In contrast to floats, satellite altimeters (such as TOPEX/POSEIDON, and the next generation Jason altimeter) provide relatively high resolution information about sea-surface height, including the oceanic eddy field. The information is complementary to that of the Argo program, because floats provide the subsurface vertical structure at relatively coarse horizontal resolution whereas altimeter data provides a higher resolution (in time and, especially, space) picture of sea-level variability. Sea surface temperature is also routinely available from satellites.

The cost of the Argo program is in excess of \$40 M, and that of the Jason and other satellite missions much higher still. Assimilating this data into ocean models, and using these models for climate prediction, greatly increases the value of the data, and will be an integral part of the proposed project. We expect that this aspect of the proposed work will be conducted in collaboration with other NOAA laboratories, in particular PMEL, AOML and NODC, and as part of the NOAA Ocean Data Observations and Services initiative. Indeed, our proposed work provides a natural and important context for this activity. R. Molinari of AOML is on the Argo Science Team and has expressed interest in collaborating with GFDL in this endeavour. Sydney Levitus of NODC, perhaps the leading provider of oceanic data sets in the country, has expressed a similar interest.

5 Climate Variability on the Decadal to Multidecadal Timescale

Over the past several years a bewildering variety of oscillations and mid-to-high-latitude hemispheric, decadal-scale phenomena have emerged. These include the Pacific Decadal Oscillation (PDO), the North Atlantic Oscillation (NAO), the Northern-Hemisphere Annular Mode (NAM), the Arctic Oscillation (AO), the Antarctic Oscillation (AAO) and the Antarctic Circumpolar Wave (ACW). See, among many others, Walker and Bliss (1932), Bjerknes (1964) and Thompson and Wallace (2000). These modes (not all independent) are in some contrast with their better-known low-latitude counterpart, the Southern Oscillation, part of the El Niño-Southern Oscillation phenomenon. Typically, the mid- and high-latitude oscillations have a much longer period and a less clear signal in terms of, say, a surface pressure index than ENSO. The modes are nevertheless potentially very important for climate variability and predictability — indeed Wallace (2000) comments that ‘This phenomenon [the NAO/NAM] rivals the El Niño Southern Oscillation (ENSO) in terms of its significance for understanding global climate variability and trends.’

The existence of any putative quasi-regular phenomena on a decadal timescale would indicate that the phenomena might in principle be predictable, and some recent numerical work is encouraging in this regard. For example, recent work by Griffies and Bryan (1997) suggests that the state of the interior North Atlantic ocean is predictable for a decade or longer. Furthermore, Rodwell *et al.* (1999) find that they are able to simulate trends in the NAO (an atmospheric index) over the past few decades by imposing the sea surface temperature and allowing the model atmosphere to respond. Taken together, these results might imply that there is useful predictability in the climate system on the decadal timescale, and that our numerical models may be close to being able to capture this predictability. Such conclusions will only hold if the evolution of oceanic SST anomalies is largely determined by the ocean’s own internal dynamics. The fully-coupled ocean-atmosphere system may be less predictable, because, for example, the evolution of SST anomalies may be largely determined by stochastic and essentially unpredictable atmospheric processes (Bretherton and Battisti 2000). Also, the trend in the AO may be due to a change in atmospheric dynamics associated with increased greenhouse gas concentrations (Shindell *et al.* 1999). Certainly, it is not clear that oscillations such as the NAO are any more than red-noise (Wunsch 1999), and the observed trends may simply be a natural fluctuation in that noise. In any case, the consequences of changes in such oscillations would be profound, and this alone dictates that a considerable effort should be undertaken to better understand and predict them. For example, Delworth and Dixon (2000) show that the recent upward trend in the AO index, leading to an increase in the westerly winds over the mid and high latitudes of the North Atlantic, could substantially delay the weakening of the thermohaline circulation that is projected as a consequence of increased high latitude precipitation in a warmer world.

Another facet of predictability is the possible decadal variability of ENSO itself. Over the past century certain decadal long periods have been virtually bereft of El Niño events (for example the 1930s), whereas in other decades their frequency and amplitude is abnormally high. Recent work indicates that variations in the thermocline depth, possibly caused by decadal scale oscillations in mid-latitudes, may be the underlying reason. With global warming however, the abyssal ocean is expected to warm more slowly than the upper ocean, the depth of the main thermocline may change as a consequence, and there may be potentially significant associated changes in El Niño frequency and intensity. This example further underscores the importance of the interaction of natural variability with anthropogenic climate change, which lies close to the heart of our proposed work.

Unfortunately, the current generation of fully-coupled ocean-atmosphere climate models are currently unable to reproduce trends in the mid-latitude oscillations themselves, nor can they properly produce El Niño events with the right magnitude and frequency. We need both better models and a better understanding of the dynamics of such oscillations in order to have confidence in our models’ ability to correctly capture and predict them; a similar plea is made also by Wallace (*op cit*). In addition to a continuing effort at GFDL to improve the quality of our GCMs, and the use of such models to study

ocean-atmosphere interaction, research activities in the Center must involve modelling and theoretical studies that allow feedbacks and oscillations to be studied in a conceptually simpler framework (e.g., Lau and Nath 1994, Saravanan *et al.* 2000). One potentially large source of predictability decay lies in the effects of eddies on the large-scale circulation of the ocean: an upscale cascade of energy from the eddy scale might provide an additional, perhaps unpredictable, source of variability to the large-scale circulation. The present (and next) generation of computers are insufficiently powerful to incorporate this effect in the global models to be used for climate predictions, so such effects must be studied using more idealized models and included parametrically in the global models, at the same time as we prepare for the future use of fully eddy-resolving ocean models in climate simulations.

6 Quantifying Natural Variability and Extreme Events

As we have stated, climate predictions will be expressed in terms of probabilities, using ensembles of integrations differing in initial conditions and other parameters. All deterministic methods can be developed for climate prediction are certainly subject to uncertainties, and even the level of uncertainty is uncertain. Thus, although we will estimate a probability density function (PDF) for the future climate state from the ensemble, we would like to have an independent check on its width.

Furthermore, many of the most significant climate impacts may involve changes in high impact events such as droughts, floods, hurricanes and so forth, so that the type of climate projections with the most economic value will likely involve the probability of such events. The possibility of local or global *rapid climate change* is another issue that demands similar attention. Prediction of such phenomena is one of the hardest tasks to ask of climate models, for it involves understanding how subtle changes in the global circulation might lead to changes in the frequency or intensity of much smaller scale phenomena, which are typically not well resolved by the global models.

Two approaches will be used to address this issue; one is primarily model-based, the other involves extensive use of historical data. In the first, we use high resolution, regional models to examine how smaller scale phenomena might change in future climates. For example, in a prototypical study, Knutson *et al.* (1998) examined how hurricane intensity might change in a warmer world, using model data from a global warming simulation to provide boundary conditions for a high resolution, hurricane model. They found hurricane intensity typically increased in a warmer climate. Similar techniques can be used to examine how the behavior of other small-scale phenomena might be modified in response to a changed large-scale circulation. The development of the powerful Flexible Modelling System (FMS) at GFDL will allow changes in model configuration and resolution to be made relatively easily, greatly facilitating the way such studies can be performed.

But even if our models were adequate to accurately simulate such small-scale phenomena, practical limitations will likely limit the number of ensemble members to a value well below that required to adequately estimate the tails of the PDF of the climate state. Thus, for our second approach, we turn to the historical record to attempt to quantify the natural variability of the climate system. One particular methodology assumes that changes in variability due to anthropogenic forcing will be small, and simply shifts the PDF based on changes in central value. To use this approach, very good quantitative estimates of the PDFs of the climatic quantities of interest are required. Considerable data are available for estimating such PDFs. During the 20th century, the instrumental climatic data available over North America and Europe are particularly good, especially in the latter half of the period, but the duration of the data is not sufficient to allow an accurate assessment of the probability of unlikely events; that is, to determine the tails of the PDF. This problem is exacerbated by the high likelihood that anthropogenic trends are present during this period, which makes it difficult to isolate the low-frequency internal variability of the climate system.

To address this issue, development of paleoclimatic proxies with annual resolution have allowed very long (up to 1000 years) climatic time series to be developed for certain sites. There are also a few long instrumental records extending back to the late 18th and early 19th centuries. Such records, whether proxy-derived or directly measured, have the promise of delivering important information about climate variability. Unfortunately, the spatial coverage of these long records is very limited. Innovative techniques (e.g., Mann and Bradley 1999) have allowed the reconstruction of large-scale temperature patterns, from climatic proxies, but the fidelity of such reconstructions is probably limited to relatively large scales. Some of the most important climate impacts result from smaller-scale anomalies, so a strategy must be developed for estimating the frequencies of such events at scales unresolved by the data.

Long climate integrations can aid in this effort, and part of our proposed activities will involve creating an ensemble of very long integrations of a climate model, for example for the last millennium, using the most accurate forcing functions possible as developed by the Atmospheric Processes Group at GFDL. This includes the effects of variations in greenhouse gases, aerosol content, volcanic activity and solar variations over the last three hundred years, with less accurate estimates of these forcings over the last one thousand years. Then, if adequately calibrated using existing paleoclimatic and long instrumental records, estimates of climate variability from models can be used to fill the spatial gaps between such long records. Because, they can be run for periods much longer than our best records, such model integrations can be used to better estimate the tails of the climate PDFs, and thus the return periods of infrequent events and the likelihood of such events occurring in the future.

7 Radiative Forcing and the Carbon Cycle

On the multi-decadal timescale, determination of the levels and the interaction of the external forcing agents becomes a key issue in quantification of climate change. These agents include the major well-mixed greenhouse gases such as CO₂ and methane, as well as sulfur and particulate carbon. On the multi-decadal to centennial timescales, as the climate system loses its memory of the initial conditions our predictions become increasingly dependent on various emissions scenarios. The human-influenced sources and sinks of greenhouse gases and aerosols become nearly impossible to predict, but this merely underlines the importance of understanding their effects in order to be able to make rational policy decisions. Indeed, future projections of greenhouse gases and aerosols are needed for at least three reasons: (i) to help quantify anthropogenic climate change; (ii) because of the potential response of terrestrial and marine biota to climate change; (iii) because the development of sensible emission and mitigation strategies is dependent on making such projections as quantitative as possible.

Carbon uptake is one significant source of uncertainty in global warming scenarios: there is considerable uncertainty about how much this uptake might vary naturally on seasonal-to-interannual to decadal timescales, and how this may be affected by increasing industrialization. In order improve such projections, we need to be able to answer the following fundamental questions:

1. Where and how large are the main current terrestrial and oceanic sources and sinks of CO₂?
2. How are these likely to change in the future, both as a result of natural variability and anthropogenic climate change?

To answer these requires in turn an understanding of how the carbon cycle and the climate system interact and function together, and this will be an important goal of the Center. Considerable progress has already been made toward this by the Carbon Modelling Consortium (CMC) at Princeton University, but understanding how the variability of the biogeochemical processes is linked to that of the physical

processes remains a key problem in predicting climate change. The interaction of other elements is even less understood, especially non well-mixed greenhouse gases and particulate matter.

Thus far, most of the assessments of the oceanic uptake of anthropogenic CO₂ have assumed that the biological system is not altered by climate change. Thus the chemical solubility of the ocean, but not potential changes in biological activity, have been included in the models. There is increasing evidence that this is unrealistic and overconstrains the models (Denman *et al.* 1996). The response of such models both to natural variability and to anthropogenic climate change may thus be too limited, potentially resulting in an poor estimate of the variability of future levels of CO₂.

Similar issues arise in the terrestrial biosphere, where the feedback between climate and vegetation could lead to human activity having very untoward effects. In the most extreme case, deforestation could lead to a dry tropics, which would then significantly feed back onto the climate system as a whole. Newly developed models of the terrestrial biosphere, based on appropriately scaled-up models of communities and ecosystems, can be used to ascertain whether such scenarios are at all tenable. However, the complete incorporation of both terrestrial and marine ecosystem models into the GCMs is really necessary to understand the full range of feedbacks between the physical and biogeochemical systems, and to obtain accurate predictions of likely future CO₂ levels.

8 Detection, Attribution and Prediction

The issues involved in ‘detection and attribution’ play a role not only in how climate has changed thus far in response to increasing greenhouse gases, but also in quantifying future changes and the plausibility of such changes. In practice, detection usually means that there is a significantly greater degree of resemblance between observations and a climate simulation with anthropogenic forcings than between observations and a simulation without anthropogenic forcings. The degree of resemblance is usually measured by a regression coefficient of observations onto a model-simulated climate change signal, say the average difference between a simulated reference climate and the simulated response of the climate system with increased concentrations of greenhouse gases and aerosols (e.g., Santer *et al.* 1995; Hegerl *et al.* 1996). This type of analysis is sometimes known as *optimal fingerprinting*. Attribution goes further and attempts to associate signals with particular causes, again via the use of model simulations and pattern matching (e.g., Hasselmann 1997). An irreducible physical difficulty in all these procedures comes from the natural variability in both the climate and the simulations. Natural variability and anthropogenic climate change are frustratingly intertwined, and no matter what statistical tests are employed, the natural variability of the system must be properly represented in the models or the answers may be misleading.

Similar issues recur in the predictability problem, because of the need to understand the degree to which the climate response is due to natural variability or to a forced response to greenhouse gases. In the standard problem (as it has been posed in the past) one simply wishes to distinguish the forced response due to greenhouse gases from that due to natural variability (‘noise’). We propose to go one step further, and to understand what aspects of the *natural variability* of the climate system are predictable, and can thus be regarded as signal rather than noise. There are two classes of questions we wish to address:

1. Which aspects of the climate system are most predictable simply in the context of global warming? That is, which aspects of the future climate system robustly differ from a climatology with no increased greenhouse gases?
2. Which aspects of the climate system can be better predicted by initializing the model from data? Conversely, what aspects of the initial field is the future evolution of the climate most sensitive to?

Key to answering these important questions will be performing multiple *ensembles* of climate integrations, initialized with differing oceanic and atmospheric states, to isolate the role of natural variability and the initial conditions. Given such an ensemble, various statistical tests may be employed to help answer these questions (e.g., Anderson and Stern 1996; Schneider and Griffies 1999). For example, we can expect the SST of the North Atlantic to be predictable, for some time period, if the initial data are given. What other fields, especially atmospheric ones, might be predictable in this sense? The quantification of such issues, indeed the quantification of climate predictability, will be a central theme of our Center. The combination of carefully designed ensemble experiments, assimilating data to determine the initial conditions, and new statistical methods, is a novel and powerful approach to this problem.

9 Research Activities: an Overview

Let us now summarize some of the research that we propose, including the numerical experiments that will be done, the subsequent analysis, and the ongoing basic research.

We expect to be continually performing multiple ensembles of experiments to examine the evolution and predictability of the climate system. For example, we will perform a set integrations which differ in their initial conditions in the ocean and sea ice, since these are the slowly varying components of the system. We intend that the ocean state should incorporate assimilated observations, with other ensemble members being perturbations around that state. Since practical issues will force the ensemble to be small, it will be important to carefully choose the ensemble both so that the initial error is representative of our lack of knowledge of the initial conditions and so that the oceanic error growth is not artificially small. Effectively, the PDF of the data error must be assimilated in a manner similar to that of the data itself.

In addition, we will form an ensemble of integrations with differing *atmospheric* initial conditions but identical oceanic initial conditions. If there is significant climate predictability, the ensemble members that differ only in initial atmospheric state should track each other better than members with differing oceanic initial conditions. Our statistical predictability analyses will quantitatively assess this, determine which components of the climate system might be predictable, and which aspects of the initial state the evolution of the system is most sensitive to. We again emphasize that for such experiments to be meaningful, climate models that plausibly simulate the natural variability of the climate system must be employed. This will entail continual model development in conjunction with existing activities at GFDL, and continual testing of the predictions of the models against suites of oceanic and atmospheric observations.

We are now actively engaged in the planning of our first set of experiments. We will take three different oceanic initial conditions corresponding to calendar year 2000 from three different transient greenhouse gas integrations. Thus, we have three different but plausible oceanic initial states. Each case will then be run for 50 years, each with three different atmospheric initial conditions, resulting in nine total experiments. A much smaller ensemble will be continued out to 100 years. We envision that this kind of activity will go on continually throughout the lifetime of the Center, using increasingly realistic initial states and increasingly more realistic models and forcing functions, and would be the basis for the assessments produced.

In parallel and coordinated with this activity will be research into the causes and nature of climate variability and the mechanisms that limit its predictability. This will entail research into the various feedback mechanisms that amplify and diminish the anthropogenic forcings, and research into the forcings themselves. This will include research into the variability of the carbon cycle and the terrestrial and marine uptake of carbon dioxide, research into the nature of oceanic variability and its influence on the atmosphere, and research into the nature and possibility of extreme climate events and rapid climate

change. Study of the historical record in conjunction with long model integration will be important in quantifying natural variability and will serve as an important check on our models.

10 Structure and Composition

The proposed Center will be composed of both GFDL and Princeton University scientists, and collaborators from other institutions. Although a very strong foundation for this activity already exists at Princeton University and GFDL, work of the scope we have proposed will require a major enhancement of the modelling and analysis groups that already exist. We expect that significant progress could be made toward our ultimate goals by the addition of about 20 scientists and support staff, who would bring important new talents to the difficult problems we have outlined. In particular, scientific teams would need to be formed or greatly strengthened in the following areas:

1. Outreach to and interface with the impacts community.
2. Ocean Data Assimilation.
3. Model/data diagnostics and dynamics, including decadal/centennial dynamics and analysis of the historical record.
4. Model development and radiative forcing, including improved schemes for aerosols and marine and terrestrial biogeochemical cycles.

Centers similar to the one we propose do exist elsewhere in the world: for example, the Hadley Centre for Climate Prediction and Research is the main center in the UK for such activities. The Hadley Centre is a little larger than the entirety of GFDL, and its portfolio explicitly includes a mandate to simulate the present climate and understand its natural variability, to understand the factors controlling climate change, and to predict global and regional climate change up to the end of the 21st century. The Max-Planck-Institute für Meteorologie in Hamburg also investigates the nature of interannual and decadal climate variability and the predictability of climate variability on interannual to decadal time scales. All of these efforts are very substantial indeed. Nevertheless, because of the combined expertise of GFDL and Princeton University and the strategy we have outlined here, we believe that we are in a position to make a unique and important contribution and become the leader in this field.

We expect that the policy aspects of climate and climate change will become increasingly important in years to come; the high-level discussions stemming from the Kyoto Protocol are one indication of this. At Princeton University, the Princeton Environmental Institute (PEI) explicitly seeks to make connections between science and policy, and we expect the Center to have considerable interaction with PEI. Indeed, PEI has explicitly stated its willingness and desire to foster such interaction, and ultimately to help build a policy side to the Center. This will also involve the Woodrow Wilson School of Public and International Affairs, whose faculty are already concerned with a variety of policy issues related to the environment and energy. The eventual establishment of a policy component within the Center will enable scientists and policymakers to interact much more closely than has been possible in the past, and consequently the policy implications of climate change to be explored in much more depth.

For a Center to embrace all of the activities we have outlined in this prospectus will require the development of a new level of infrastructure between GFDL and the University. One possible avenue for this is the development of a Joint Institute, but the development of any such infrastructure must be done carefully and thoughtfully.

References

- Anderson, J. and W. F. Stern, 1996. Evaluating the potential predictive utility of ensemble forecasts. *J. Climate*, **9**, 260–269.
- Anderson, J. H. van den Dool, A. Barnston, W. Chen, W. Stern and J. Ploshay 1999. Present-day capabilities of numerical and statistical models for atmospheric extratropical seasonal simulation and prediction. *Bull. Am. Meteor. Soc.*, **80**, 1349–1361.
- Argo Science Team, 1998. On the design and implementation of Argo: an initial plan for a global array of profiling floats. International CLIVAR Project Office Report 21, GODAE Report 5. GODAE International Project Office, Melbourne, Australia, 32pp.
- Argo Science Team, 2000. Report of the Argo Science Team Meeting (AST 2) Southampton, March 2000. 35pp.
- Bjerknes, J., 1964. Atlantic air-sea interaction. *Adv. Geophys.*, **10**, 1–82.
- Bretherton, C. S. and D. S. Battisti, 2000. An interpretation of the results from atmospheric general circulation models forced by the time history of the observed sea surface temperature distribution. *Geophys. Res. Lett.*, (in press).
- Davis, R. E., 1998. Preliminary results from directly measured middepth circulation in the tropical and south Pacific. *J. Geophys. Res.*, **103** 24619–24639.
- Delworth, T. and K. Dixon. 2000. Implications of the recent trend in the Arctic Oscillation of ocean circulation and climate change. *J. Climate*, (in press).
- Delworth, T. and T. Knutson. 2000. Simulation of early 20th century global warming. *Science*, **287**, 2246–2250.
- Denman, K., E. Hofmann and H. Marchant, 1996. Marine biotic responses to environmental change and feedbacks to climate. In *Climate Change 1995: The Science of Climate Change*, Contribution of Working Group 1 to the Second Assessment Report of the IPCC, ed. J. T. Houghton *et al.*, pp. 193–227. Cambridge University Press.
- Hasselmann, K., 1997. Multi-pattern fingerprint method for detection and attribution of climate change. *Climate Dyn.*, **13**, 601–611.
- Hegerl, G. C., H. von Storch, K. Hasselmann, B. D. Santer, U. Cubasch, and P. D. Jones, 1996. Detecting greenhouse-gas-induced climate change with an optimal fingerprint method. *J. Climate*, **9**, 2281–2306.
- Griffies, S. M. and K. Bryan, 1997. Predictability of North Atlantic multidecadal variability. *Science*, **275**, 181–184.
- Lau, N-C and M. J. Nath 1994. A modeling study of the relative roles of tropical and extratropical SST anomalies in the variability of the global atmosphere-ocean system. *J. Climate*, **7**, 1184–1207.
- Rodwell, M. J., D. P Rowell and C. K. Folland, 1999. Oceanic forcing of the wintertime North Atlantic Oscillation and European climate. *Nature*, **398**, 320–323.
- Mann, M. E., and R. S. Bradley, 1999. Northern Hemisphere temperatures during the past millenium: inferences, uncertainties, and limitations. *Geophys. Res. Lett.*, **26**, 759–762.

- Santer, B. B., T. M. L. Wigley, T. P. Barnett and E. Anyamba, 1996. Detection of climate change and attribution of causes. In *Climate Change 1995: The Science of Climate Change*, Contribution of Working Group 1 to the Second Assessment Report of the IPPC, ed. J. T. Houghton *et al.*, pp. 407–443. Cambridge University Press.
- Saravanan, R., G. Danabasoglu, S. C. Doney, J. C. McWilliams, 2000. Decadal Variability and Predictability in the Midlatitude Ocean–Atmosphere System. *J. Climate*, **13**, 1073–1097.
- Schneider T. and S. M. Griffies, 1999. A conceptual framework for predictability studies. *J. Climate*, **12**, 3133–3155.
- Shindell, D.T., Miller, R.L., Schmidt, G.A., and L. Pandolfo, 1999. Simulation of recent northern hemisphere winter climate trends by greenhouse-gas forcing. *Nature*, **399**, 452–455.
- Stern, W. F. and K. Miyakoda, 1995. The feasibility of seasonal forecasts inferred from multiple GCM simulations. *J. Climate*, **8**, 1071–1085.
- Thompson, D. W. J. and J. M. Wallace, 2000. Annular modes in the extratropical circulation. Part I: Month-to-month variability. *J. Climate*, **13**, 1000–1016.
- Wallace, J. M. 2000. North Atlantic Oscillation/annular mode; Two paradigms — one phenomenon. *Quart. J. Roy. Meteor. Soc.*, **126**, 791–805.
- Walker, G. T. and E. W. Bliss, 1932. World Weather V. *Mem. R. Meteorol. Soc.* **4**, 53–83.
- Wunsch, Carl, 1999. The interpretation of short climate records, with comments on the North Atlantic and Southern Oscillations. *Bull. Am. Meteor. Soc.* **80**, 245–256.

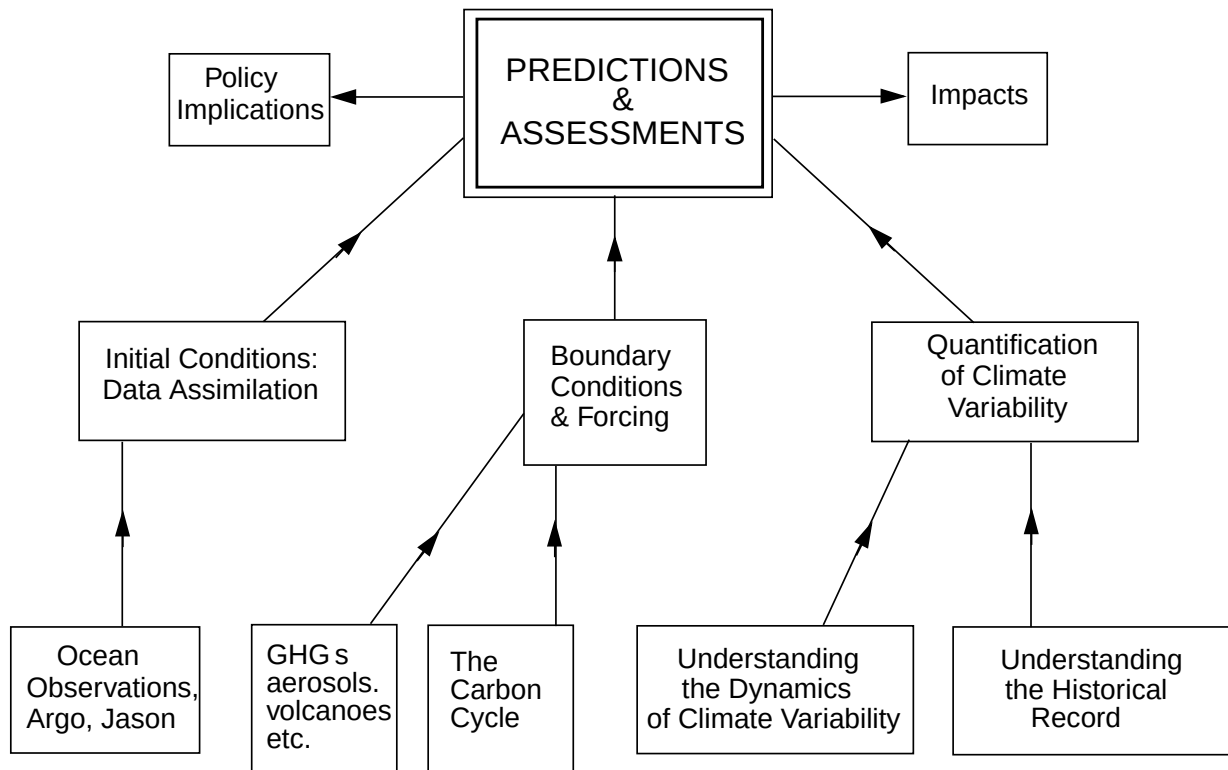


Figure 1: Schematic of some of the main activities in the proposed Center.

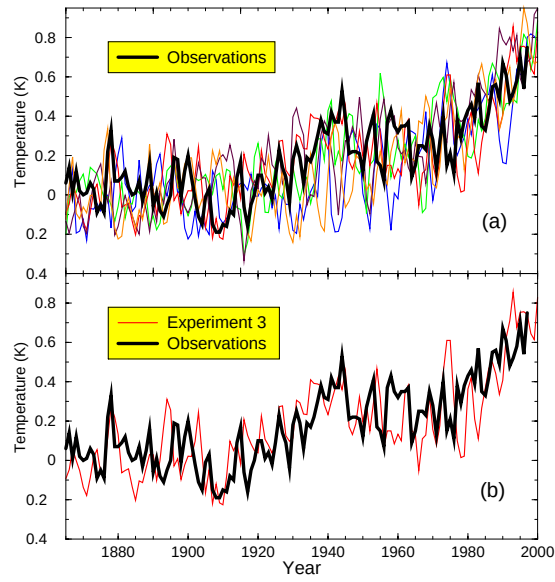


Figure 2: (a) Time series of global mean surface temperature from observations (heavy black line) and the five model experiments (colored lines). (b) Same as (a), except that only one of the model timeseries ('ensemble member 3') is shown (red line).

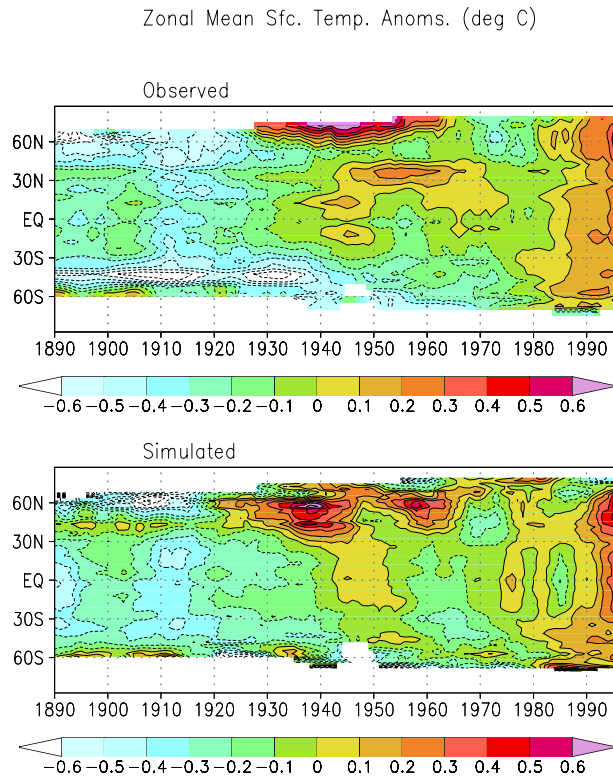


Figure 3: Zonal mean anomalies of surface temperature for the observations (upper panel) and for ensemble member 3 of fig. 2, subject to a 10-year low-pass filter.